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Mathematical morphology

1. MORPHOLOGICAL SEPARATION, RESTORATION, AND BOUNDARY EXTRACTION

To start with this exercise we are going to do a recap of the morphological operations we have seen in the lectures, and then we will apply them to the image of Evarist and Ermessenda.

1.1. Linear filters

The operations we are going to use are also linear filters. They are based on the use of a structuring element, which is a small shape that is used to probe the image. The structuring element is moved across the image, and at each position, it is compared with the underlying pixels. Depending on the operation, the output pixel value is determined by the comparison between the structuring element and the underlying pixels.

1.1.1. Dilation

$$\delta_{B(X)} = X + B = \{x + y : x \in X, y \in B\} = \cup_{y \in B} (X + y) \tag{1}$$

The first linear filter is dilation (Equation 1), which is used to grow the boundaries of objects in a binary image. The structuring element is moved across the image, and at each position, if any of the underlying pixels are 1 (white), the output pixel is set to 1. This operation can be used to fill small holes in objects or to connect nearby objects. In Figure 1 we can see the result of applying dilation to the image of Evarist and Ermessenda.

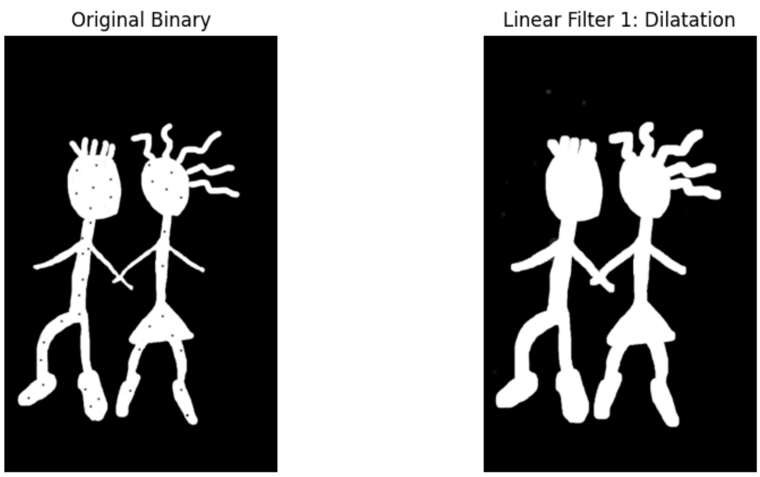

Figure 1: Dilation example.

1.1.2. Erosion

$$\varepsilon_{B(X)} = \{x \in X : x + B \subseteq X\} = \cap_{y \in B} (X - y) \tag{2}$$

The second linear filter is erosion (Equation 2), which is used to shrink the boundaries of objects in a binary image. The structuring element is moved across the image, and at each position, if all of the underlying pixels are 1 (white), the output pixel is set to 1. This operation can be used to remove small objects or to separate connected objects. In Figure 2 we can see the result of applying erosion to the image of Evarist and Ermessenda.

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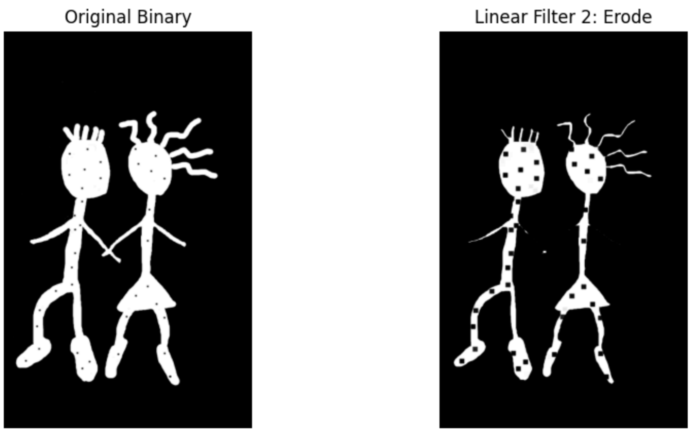

Figure 2: Erosion example.

Here is where the things get interesting, because the following filters, might help us to solve the tasks of this exercise.

1.1.3. Opening & No Holding Hands

$$\gamma_{B(X)} = \delta_{B\left(\varepsilon_{B(X)}\right)} \tag{3}$$

The third linear filter is opening (Equation 3), which is a combination of erosion followed by dilation. This operation can be used to remove small objects or to separate connected objects while preserving the shape of larger objects. In Figure 3 we can see the result of applying opening to the image of Evarist and Ermessenda. In this case, we can see that the hands of Evarist and Ermessenda are separated, but the rest of their bodies are preserved. This is because the opening operation removes small objects (the hands) while preserving larger objects (the bodies). In detail, the left arm of Evarist and both arms of Ermessenda are removed, the aim was simply to separate the hands, but the structuring element used for the opening operation was too large, which caused the removal of the arms as well. This is a common issue when using morphological operations, and it is important to choose the right structuring element size to achieve the desired result. After several trials, we found that a structuring element of size 5x5 and setting the number of iterations to 1 was the best option.

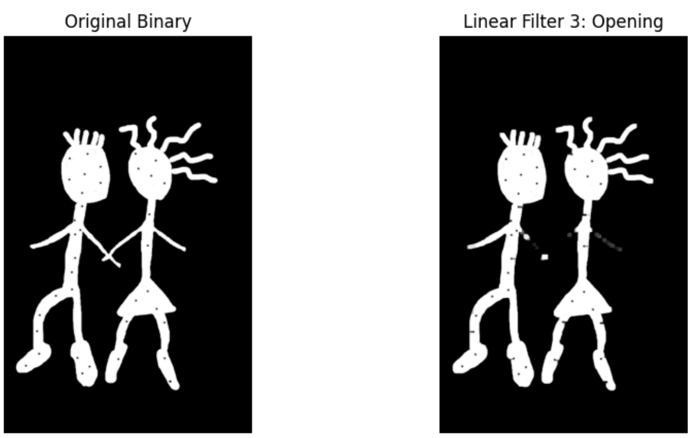

Figure 3: Opening example.

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1.1.4. Closing & Healing Holes

$$\varphi_{B(X)} = \varepsilon_{B\left(\delta_{B(X)}\right)} \tag{4}$$

The fourth linear filter is closing (Equation 4), which is a combination of dilation followed by erosion. This operation can be used to fill small holes in objects or to connect nearby objects while preserving the shape of larger objects. In Figure 4 we can see the result of applying closing to the image of Evarist and Ermessenda. In this case, we can see that the holes in the bodies of Evarist and Ermessenda are filled, but the rest of their bodies are preserved. This is because the closing operation fills small holes (the holes in the bodies) while preserving larger objects (the bodies). In this task we got a better result that we did in the last one, and finding the right structuring element size was easier, we found that a structuring element of size 4x4 and setting the number of iterations to 1 was the best option.

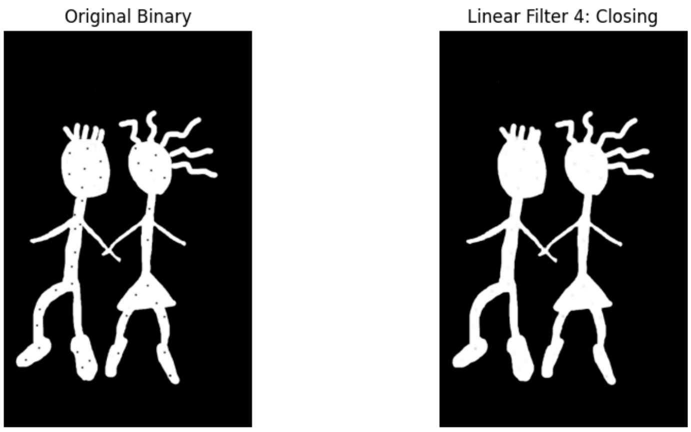

Figure 4: Closing example.

The exercise has not finished with this last filter because we still have to get the boundaries of the holding-hands couple. This goal will require the use of both dilation and erosion, but we will see how to combine them to get the desired residue.

1.2. Residues

1.2.1. Centered Gradient & Boundary Extraction

$$\nabla^c_{B(u)} = \delta_{B(u)} - \varepsilon_{B(u)} \tag{5}$$

The name of this filter (Equation 5) is quite self-explanatory, it is a gradient that is centered in the original image, and it is obtained by subtracting the erosion from the dilation. This operation can be used to edge detection in greyscale images. In Figure 5 we can see the result of applying the centered gradient to the image of Evarist and Ermessenda. For this result, we can see that the boundaries of Evarist and Ermessenda are preserved, but the rest of their bodies are extracted. The reason why this happens is because the centered gradient operation preserves the edges of objects while removing the interior pixels. This is because the dilation operation expands the boundaries of objects, while the erosion operation shrinks the boundaries of objects. When we subtract the erosion from the dilation, we are left with only the edges of objects, which is what we wanted to achieve in this task. An interesting observation out of this result is that the holes in the bodies of Evarist and Ermessenda are also preserved, this is because the dilation operation fills the holes, while the erosion operation removes the holes. When we subtract the erosion from the dilation, we are left

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with only the edges of objects, which includes the holes in this case. For this task we found that a structuring element of size 3x3 and setting the number of iterations to 1 was the best option.

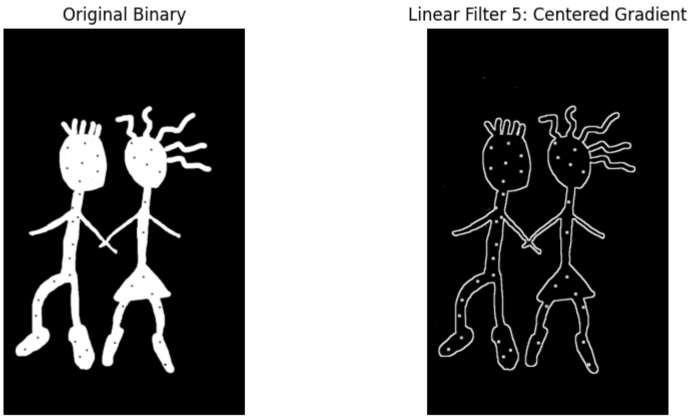

Figure 5: Centered gradient example.

1.3. Results

At the end of this exercise, Figure 6 shows the final results of applying the morphological operations to the image of Evarist and Ermessenda. We can see that the hands of Evarist and Ermessenda are separated, the holes in their bodies are filled, and the boundaries of their bodies are preserved.

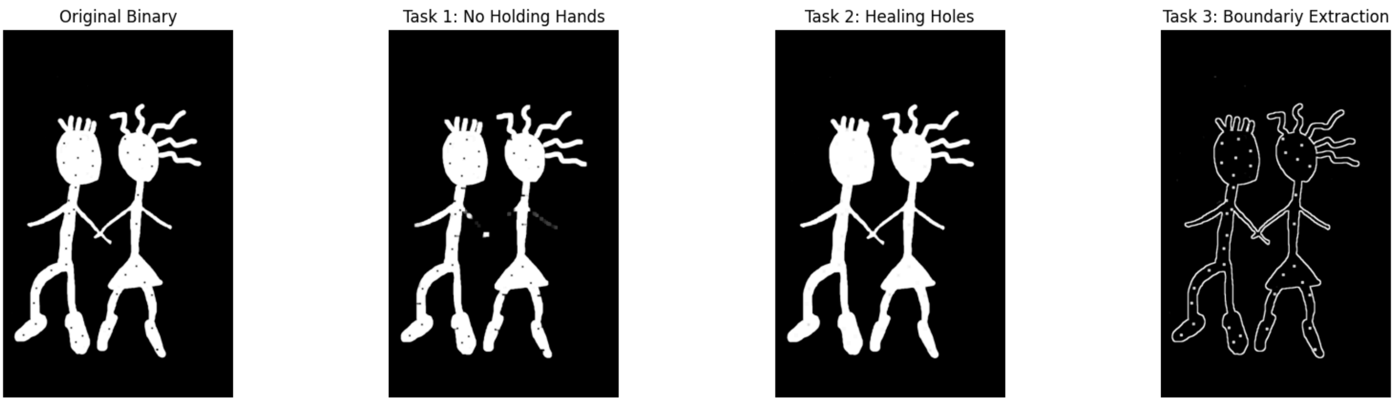

Figure 6: Exercise 1 results.

2. SEGMENTATION OF LETTERS

In this second exercise, we face a classic image processing challenge: segmenting handwritten text from an image with severe, uneven illumination. As seen in the original image, there is a strong shadow gradient from dark on the left to bright on the right. Because of this gradient, a standard global threshold would fail; a cutoff value that successfully isolates the text on the bright right side would turn the dark left side entirely black. To solve this, we will apply mathematical morphology to estimate and subtract the background.

2.1. Background Estimation via Closing

Since the text (ink) is strictly darker than the background (paper), we can use the morphological closing operation to eliminate the text and isolate the illumination gradient.

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$$\varphi_{B(X)} = \varepsilon_{B\left(\delta_{B(X)}\right)}$$

The closing filter (Equation 5), which consists of a dilation followed by an erosion, is ideal for this task. By choosing a structuring element B (in our code, a 15x15 square kernel) that is slightly larger than the stroke width of the letters but smaller than the gaps between the lines of text, we can manipulate the image features.

Because the text is dark, the initial dilation shrinks the dark ink until it completely disappears. The subsequent erosion then restores the remaining bright background features to their original spatial boundaries. The result is a smooth image containing **only** the shadow gradient, with the handwritten text effectively erased.

2.2. Illumination Correction and Binarization

Once we have an accurate mathematical estimation of the background, we can correct the uneven lighting in the original image.

2.2.1. Subtraction (Uniform Illumination)

By subtracting the original image from our estimated background, we effectively flatten the lighting across the whole image. The mathematical difference between the background and itself is near zero (black), while subtracting the dark text from the bright estimated background results in positive values (bright). This step successfully isolates the ink, leaving us with bright text on a uniformly dark background.

2.2.2. Thresholding and Inversion

Because the illumination is now perfectly uniform across the entire document, we no longer need complex adaptive algorithms. A simple, hard-coded global binarization (using a low threshold like 0 or 15) successfully separates the text from the flattened background. Finally, to restore the natural appearance of the document (dark text on a light background), we apply a bitwise inversion to flip the colors of the segmented mask.

2.3. Results

Figure 7 displays the complete step-by-step pipeline achieved by the code. The top-left shows the original image with its severe gradient. The top-right shows the isolated background estimated via the closing filter. The bottom-left demonstrates the uniform illumination achieved through subtraction, and the bottom-right presents the final, cleanly segmented text.

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Figure 7: Exercise 2 pipeline: Original, Background Estimation (Closing), Subtraction, and Final Segmentation.

3. SEGMENTATION OF NOISY LETTERS

4. MORPHOLOGICAL GRADIENTS, TOP-HAT AND BOTTOM-HAT